

AP EAMCET AGRICULTURE AND MEDICINE

Date : 24th April 2019

Number of Questions :160

Shift – II

1. A, B and C are heterotrophic plants. 'A' has no sex organs but produces sexual spores exogenously. 'B' produces sexual spores endogenously and 'C' has no sexual stage but reproduces asexually. Based on these characters, identify A, B and C respectively.
 - a. Rhizopus, Aspergillus, Alternaria
 - b. Colletotrichum, Albbugo, Penicillium
 - c. Ustilago, Neurospora, Trichoderma
 - d. Aspergillus, Agaricus, Alternaria
2. Choose the correct pair of taxa showing following sets of characters in sequence.
 - i) Tracheophytic heterosporous plants with siphonogamous oogamy
 - ii) Tracheophytic heterosporous plants with archegonia
 - iii) Archegoniates with zooidogamous oogamy

- A) Mearns and Selaginella
B) Salvinia and Pinus
C) Polytrichum and Eucalyptus
D) Cuscuta and Eucalyptus

- a. D, B, A
b. C, D, B
c. B, D, A
d. D, A, C

3. Match the following.

List-I

Scientist

A) Stephen Hales

B) J. B. Sumner

C) Camerarius

D) Haeckel

List-II

Period of Discovery

i) 20th Century

ii) 19th Century

iii) 18th Century

iv) 17th Century

List-III

Discovery

I) Root

pressure

II) Sexual reproduction in plants

III) Ecology

IV) Urease

The correct answer is

- | | | | |
|-----------|-------|----------|---------|
| A | B | C | D |
| a. ii, II | i, IV | iii, III | iv, I |
| A | B | C | D |
| b. iii, I | i, IV | iv, II | ii, III |
| A | B | C | D |

c. ii, II	i,IV	iv,II	iii,III
A	B	C	D
d. iii,I	iv,II	i,IV	ii,III

4. Choose the correct statement.

- Red colour of the red sea is due to dinoflagellates.
- Heterocysts of cyanobacteria fix atmospheric nitrogen.
- Mycoplasmas are pleomorphic due to the presence of cell wall.
- Presence of mycolic acid in the cell wall of diatoms is the reason for the left over cell deposits in large amounts in their habitats.

5. Match the following.

List-I	List-II	List-III
A) Acropetal arrangement	i) Dichasial cyme and monochasial cyme	I) Mangifera
B) Margin of thalamus enclosed the ovary	ii) Inferior ovary	II) Cypsella

C) Single ovule attached to the base of ovary	Simple raceme	III) False whorl
D) Lamiaceae	iv) Persistent pappus like calyx	IV) Guava

The correct answer is

- | | | | |
|------------|--------|---------|--------|
| A | B | C | D |
| a. iii, I | ii, IV | iv, II | i, III |
| A | B | C | D |
| b. iii, I | iv, IV | ii, III | i, II |
| A | B | C | D |
| c. ii, IV | iii, I | iv, II | i, III |
| A | B | C | D |
| d. iv, III | i, II | iii, I | ii, IV |

6. Read the following statements and identify the correct pairs of plants

- (A) modified protective bract and edible pedicel
- (B) food storing floral bud and leaf bases
- (C) photosynthetic stem and petiole.

- a. Oryza and Cocos: Aloe and Agave, Nepenthes and Australian acacia
- b. Colocasia and Apple: Dioscorea and Opuntia:

Australian acacia and Euphorbia

c. Musa and Ficus; Dioscorea and Bravophyllum:

Australian acacia and Bougainvillea

d. Cocos and Anacardium: Agave and Allium:

Casuarina and Australian acacia

7. Identify the correct ratio of haploid, diploid and triploid conditions of cells and tissues listed in the given table.

Pollengrain	Nucellus	Sporogenous tissue
Endosperm	Perisperm	Egg cell
Synergid	Zygote	Scutellum

- A. 5:3:1
- B. 3:5:1
- C. 4:4:1
- D. 2:6:1.

8. A total of 168 pollengrains with 14 chromosomes in each pollen grain are released from a mature microsporangium. Mention the correct ratio of pollen mother cells, generative cells, vegetative cells and male gametes produced by the microsporangium.

- a. 1:1:1:4

b. 1:2:1:4

c. 1:4:4:8

d. 2:1:1:4

9. Choose the correct matching.

List-I

A) Dioecious plant with Archegoniophore

B) Monoecious plant with Oogonium

C) Homothallic plants

D) Dioecious plants with pistillate flowers

List-II

I) Papaya

II) Chara

III) Fungi

IV) Maize

V) Marchantia

The correct answer is

A B C D

a. V II III I

A B C D

b. V III II I

A B C D

c. V IV III I

A B C D

d. V I II III

10. Choose correct pair from the given below related to the study of fertilization.

A	Aquatic algae	Large no. of gametes released into the surrounding	External syngamy
B	Moss plants	Large no. of gametes released into air	External syngamy
C	Seed plants	Non motile male gamete reach the egg	Internal syngamy
D	Pteridophytes	Motile male gamete reach the egg	Internal syngamy

- a. B, C, D
- b. A, B, C
- c. A, C, D
- d. A, B, D

11. Match the following.

List-I	List-II	List-III
A) Radical leaves	i) Anterior odd sepal	I) Protandrous
B) Exstipulate leaf	ii) Artificial system of classification	II) Persistent Calyx
C) Monocarpellary unilocular gynoecium	iii) Allium	III) Geocarp
D) Species plantarum	iv) Solid or hollow stem	IV) Linnaeus

The correct answer is

- | | | | |
|------------|---------|--------|--------|
| A | B | C | D |
| a. iii, I | iv, II | i, III | ii, IV |
| A | B | C | D |
| b. iv, I | iii, II | i, III | ii, IV |
| A | B | C | D |
| c. iii, II | iv, III | i, I | ii, IV |
| A | B | C | D |
| d. iii, I | iv, III | i, III | ii, IV |

12. Study the following statements.

I. Circular naked DNA is the genetic material.

II. Membrane bound cell organelles are present.

III. Cell membrane infoldings called mesosomes present.

IV. Nuclear membrane absent.

Arrange the given organisms based on their above respective characters and give the ratio

A) Bacillus

B) Nostoc

C) Noctiluca

D) Chlorella

E) Rhodospirillum

F) Trichodesmium

G) Amoeba

H) Euglena

A. 4:3:4:4

B. 4:4:3:4

C. 2:2:2:1

D. 2:2:1:2

13. Assertion (A) : Syncytium is the result of karyokinesis not followed by cytokinesis.

Reason (R) : In plant cell, middle lamella formation starts from centre and grows outwards of the cell and leads to cell wall formation.

- a. (A) and (R) are true and (R) is the correct explanation of (A).
- b. (A) and (R) are true. but (R) is not the correct explanation of (A).
- c. (A) is true (R) is false.
- d. (A) is false. (R) is true.

14. Identify the correct series of biomolecules which donot belong to the given category

- Homopolymer
 - Secondary metabolite
 - Neutral Aminoacid
 - Aromatic anunoacid
- a. Inulin. Rubber. Valine. Tryptophan
 - b. Protein. Scent, Tyrosine, Tryptophan
 - c. Cellulose, Protein, Glutamic acid, Tyrosine
 - d. Rubisco, Sugar, Lysine. Valine

15. Observe the following functions of the cell.

Synthesis of protein and enzymes
Synthesis of lipids
Synthesis of Glycolipids

- Enzymes optimally active in acidic pH.

Arrange the given cell organelles with their respective functions.

- a) Golgi apparatus
- b) Rough endoplasmic reticulum
- c) Smooth endoplasmic reticulum
- d) Lysosome
- e) Ribosome
- f) Mitochondria

- a. b, a, c, e
- b. e, a, c, d
- c. b, c, r, f
- d. e, c, a, d

16. Assertion (A) : Ions and organic material concentration is higher in cytoplasm than vacuole.
Reason (R) : Tonoplast facilitates the transport of ions and other materials against concentration gradient into the vacuole.

- a. (A) and (R) are correct. (R) is the correct explanation to (A).
- b. (A) and (R) are correct. (R) is not the correct explanation to (A).
- c. (A) is true. (R) is false.
- d. (A) is false. (R) is true.

17. Arrange the following aspect of a double stranded DNA molecule with the 3'AGT CGT CGAA5' sequence in ascending order
- A) Number of phosphodiester bond
 - B) Number of bonds
 - C) Number of total pyrimidines
- a. A, B, C
 - b. A, B
 - c. B,A,C
 - d. C, B, A
18. The tissues from which cambium ring is originated in dicot root and dicot stem.
- a. Cambium between xylem and phloem : Pericycle and medullary rays
 - b. Tissues below phloem bundles and portion of pericycle tissue : Medullary rays, Cambium and phellogen.
 - c. Cambium. pericycle tissue : cambium and phellogen
 - d. Pericycle tissues : Cambium, Medullary rays
19. Identify the correct series of tissues which show the following features.
- Maintaining pressure gradient in the sieve tubes.

- Tissues absent in monocots.
- Mature cells lacking nucleus.
- Dead cells found in the fruit pulp of sapota.
- a. Phloem fibres. Intercalary meristem, Albuminous cells. Collenchyma
- b. Phloem parenchyma, Phloem. Companion cells. Cork cells
- c. Companion cells. Phloem parenchyma. Sieve tubes, Sclereids
- d. Companion cells. Sievetubes. Albtuninous cells. Sclereids

20. Assertion (A) : A mature leafy tree produces as much oxygen in a season as 10 People inhale in a year.
Reason (R) : Forests provide a vast bank for CO₂ and a huge amount of CO₂ is deposited in timber.
- a. (A) and (R.) are true and (R) is the correct explanation of (A).
 - b. (A) and (R) are true but (R) is not the correct explanation of (A).
 - c. (A) is true. (R) is false.
 - d. (A) is false. (R) is true.

21. Identify the correct phenomenon with reference to transport in plants.
- a. Hormones are chemical stimulators always transport towards growing points.
 - b. Transport of water and minerals through non living conducting tissue is multidirectional.
 - c. Transport of solutes through living conducting tissue is unidirectional.
 - d. Diffusion and cytoplasmic streaming supplemented by active transport is necessary for movement of substances to shorter distance.
22. Deficiency of a set of three mineral ions induced by excessive absorption of manganese lead to the following set of metabolic disorders. Identify the set of deficit mineral ions.
- A) Transfer of electrons by ferredoxin and Cytochromes
 - B) Synthesis of DNA and RNA
 - C) Normal functioning of the cell membrane
- a. Identify the set of deficit mineral ions.
 - b. Iron. Magnesium and Nickel
 - c. Iron. Molybdenum and Magnesium
 - d. Iron. Magnesium and Calcium

23. Choose the correct combination.

Enzyme reaction - Enzyme name - Enzyme class

A) Glutarnic acid $\rightarrow$ Glutamine - Glutamate synthetase - Ligase

B) Fru 1, 6 BP $\rightarrow$ Frn 6 M.P. - Phosphatase - Hydrolase

C) Glucose $\rightarrow$ Glu 6 P - Hexokinase - Transferase

D) Argino acid $\rightarrow$ Arginine - Argino succinase – Lyase

a. A, B, C, D

b. B, C

c. C, D

d. A, B

24. Which of the following statement is correct with reference to absorption spectra and action spectra

A) In blue light. Chl. a shows maximum absorption peak than Chi. b

B) In Red light. Cll. a shows maximum absorption than Chl. b

C) Absorption spectrum of chlorophyll is maximum in blue light, wheras action spectrum is maximum in red light.

D) Rate of photosynthesis is measured by O₂ release.
The correct answer is

- a. A, B, C
- b. A, B
- c. A, B, C
- d. B, C, D

25. Identify the wrong statements with respect to ATP synthesis in Chloroplast based on Chemiosmotic hypothesis

- A) Proton gradient across the thylakoid membrane increases due to Quinone cycle
- B) The movement of protons through F_0 leads to decreased pH in stroma of Chloroplast
- C) Breakdown of proton gradient is a cause for release of energy
- D) Splitting of water molecules produce protons in stroma of chloroplasts

- a. A, B
- b. B, C
- c. B, D
- d. A, D

26. Arrange the following steps in ascending order based on no. of ATP molecules gained when one glucose molecule participate in aerobic respiration.

- I) Total no. of ATP in Krebs cycle

II) Succinyl CO A $\rightarrow$ Succinic acid

III) Pyretic acid $\rightarrow$ Acetyl CO A

IV) PGAL $\rightarrow$ BPGA

27. Based on plant growth regulation ability choose the correct combinations.

A) Carotenoid - ABA - Tolerance of stress

B) Terpene IBA Root elongation

C) Adenine derivative - KN Production of new leaves

D) Purine derivative - C_2H_4 - Fruit ripening

a. II, IV, III, I

b. I, II, III, IV

c. I, IV III, II

d. II, III, I, IV

28. Choose the correct mathematical expressions of linear growth and sigmoid growth and no. of progeny undergo division in both growth curves.

a. $L_t = L_o + rt$; $W_t = W_o e^{rt}$; 1, 2

b. $L_t = L_o$; $W_t = W_o$; 1, 1

c. $L_t = L_o + rt$; $W_1 = W_o e^{rt}$; 2, 1

d. $L_t = L_o + ro$; $W_1 = W_o e^t$; 2, 2

29. Match the following.

List-I	List-II	List-III
A) Photoautotrophs	i) Oxidation of inorganic substances	I) Bacillus
B) Photoheterotrophs	ii) Organic detritus	II) Beggiotoa
C) Chemoautotrophs	iii) Organic CO ₂	III) Chromatium
D) Saprophytes	iv) Atmospheric CO ₂	IV) Rhodospirillum

The correct answer is

- A B C D
 a. iv, III iii, IV i,II ii,I
 A B C D
 b. iv, III i,II iii,IV ii,I
 A B C D
 c. iv, I i,iv iii,II ii,III
 A B C D
 d. iv, IV iii,I ii,III I,II

30. Assertion (A) : Enveloped viruses like measles. Rubella and Adenovirus attached to the susceptible host by spikes.

Reason (R) : Influenza virus is an enveloped virus attached to susceptible host by spikes.

- a. Both (A) and (R) are true and (R) is the correct explanation of (A).
- b. Both (A) and (R) are true but (R) is not the correct explanation of (A).
- c. (A) is true, but (R) is false.
- d. (A) is false. but (R) is true.

31. Find the incorrect statement.

A) Chromosomal alternations are caused either by deletion or gain.

B) Genes with low recombination show linkage whereas loosely arranged genes show higher recombination.

C) Frequency of recombinations between gene pairs on the same chromosome indicates their activity.

D) Segregation of genes occur in a less frequency in sexual reproduction.

- a. A, B
- b. C, D
- c. B, C
- d. A, D

32. Identify the true statements regarding a Double standard DNA molecules of length 170 A and possess purine and pyrimidines in a ratio of 40% and 60% on one strand.

A) No. of Hydrogen bonds 130.

B) No. of phosphodiester bonds in one strand are 49.

C) No. of dicyclic rings are 20. D) No. of thymines are 20.

a. A, B, C

b. B, C, D

c. A, B, D

d. A, C, D

33. Identify the correct statement.

a. Capping of lm RNA means the addition of adenylate residues at 5' end.

b. y gene of lac operon encodes transacetylase enzyme.

c. in Eukaryotes structural genes are interrupted by coding sequences.

d. The lac operon has two regulatory genes and three structural genes.

34. A student isolated a new virus which is able to infect E.coli. The following experiment has to follow to confirm the hereditary material of new virus.

- Hershy Chase experiment — Growth on radioactive Phosphorous medium — infection to E.coli.
- Mathew Meselon and Franklin Stahl experiment — $^{15}\text{NH}_4\text{Cl}$. ^{14}N medium -centrifugation.
- Mathew Meselson and Franklin Stahl experiment — Digestion with protease - CsCl_2 gradient centrifugation
- Hershy Chase experiment — Growth on radioactive sulphur medium — infection to E.coli.

35. Match the following.

List-I	List-II	List-III
A) Hind III	i) Agarose gel	I) Six base pairs
B) pBR ³²²	ii) Avv bacteriunt	II) Selectable marker
C) T-DNA	iii) Ampicillin	III) Elution
D) DNA	iv) Recognition sequence	IV) Trausgenic plant

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A	B	C	D
a. iv, I	i,III	iii,II	ii,IV
A	B	C	D
b. iv, I	iii,II	i,III	ii,IV
A	B	C	D
c. iv, I	iii,II	ii,IV	i,III
A	B	C	D
d. i,III	ii,IV	iii,II	iv,I

36. A, B and C are transformed host cells with r-DNA made of P^{BR322} 'A' has foreign DNA insert at Bam HI site. 'B' has foreign DNA insert at p_{uvl} and 'C' has DNA insert in Lac 2. When they are grown on media for selection following changes are observed.

- A) 'C' transformant produces blue coloured colonies
- B) 'C' transformant colonies do not produce blue colour
- C) 'A' transformant survive on the medium containing ampicillin and 'B' survive on tetracyclin containing medium
- D) Due to insertional inactivation, 'A' survive on Tetracyclin containing medium and 'B' survive on medium with ampicillin

Identify incorrect statements

- a. A, D
- b. B, D
- c. B, C, D
- d. A, C

37. Choose the correct sequence of steps in PCR

- a. Annealing - Denaturation - Extension - Amplification
- b. Amplification - Annealing - Denaturation - Extension
- c. Denaturation - Annealing - Extension - Amplification
- d. Denaturation - Amplification - Extension - Annealing

38. Assertion (A) : Blood cholesterol is lowered with statins produced by yeast.

Reason (R) : Statins inhibit synthesis of cholesterol through competitive inhibition.

- a. Both (A) and (R) are true and (R) is the correct explanation of (A).
- b. Both (A) and (R) are true but (R) is not the correct explanation of (A).
- c. (A) is true, but (R) is false.

d. (A) is false. but (R) is true.

39. Match the following.

List-I	List-II	List-III
A) Biological nitrogen fixation	i) Ti ichodenna	I) Bacterium
B) Mycorrhiza	ii) Propionibactenum	II) Free living fungus
C) Biocontrol	iii) Azospirillum	III) Fungus
D) Swiss cheese	iv) Glomus	IV) Free living bacterium

- | | | | |
|------------|---------|---------|---------|
| A | B | C | D |
| a. iii, IV | iv, III | ii, I | I, II |
| A | B | C | D |
| b. iii, IV | iv, III | i, II | ii, I |
| A | B | C | D |
| c. iii, IV | ii, II | iv, III | i, I |
| A | B | C | D |
| d. ii, I | iv, II | i, II | iii, IV |

40. Choose the correct statements.

- I) IRS is a semi dwarf variety of rice developed in India.
 - II) Hybrid sugarcane show high sugar and high yield.
 - III) Hybrid millets are resistant to water stress.
 - IV) Sonalika is a high yielding and disease resistant hybrid variety.
- a. I,II,III
 - b. II,III,IV
 - c. I,III,IV
 - d. I,II,IV

41. Assertion (A) : The tropical Amazon rain forest in South America has the greatest Biodiversity on the earth

Reason (R) : The tropical latitudes have long evolutionary time

- a. Both (A) and (R) are correct and (R) is the correct explanation of (A)
- b. Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- c. (A) is correct but (R) is not correct
- d. (A) is not correct but (R) is correct

42. Assertion (A) : In old age. bones become more brittle
Reason (R): During ageing. the quantity of organic matter increases in a bone

- a. Both (A) and (R) are correct and (R) is the correct explanation of (A)
- b. Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- c. (A) is correct but (R) is not correct
- d. (A) is not correct but (R) is correct

43. Study the following statements

- A) Triploblastic animals are usually biradially symmetrical
- B) Certain organs in vertebrates are covered by parietal peritoneum only on their ventral side.
- C) Pancreas is an example for holocrine gland
- D) Elastic cartilage is found in the wall of Eustachian tubes

Among the above identify the correct statements

- a. A, B
- b. B, C
- c. C, D
- d. B, D

44. Match the following

List – I

- A) Cuttle Fish
- B) Devil Fish
- C) Cray Fish BD
- D) Jelly Fish

List - II

- I) Green glands
- II) Medusoid form
- III) External shell
- IV) Internal shell
- V) Shell is absent

The correct answer is

A B C D

a. III V II I

A B C D

b. IV V II I

A B C D

c. IV V I II

A B C D

d. III II I V

45. Study the following regarding arthropods

- A) In some spiders respiration is performed by both book lungs and trachea
- B) Development of king crab includes trilobite larva
- C) Blood of arachnids contain haemocyanin

D) Tin diplopods mandibles are modified into gnathochilaritun

From the above. Identify the incorrect statement(s)

- a. A & C
- b. B & D
- c. C
- d. D

46. Teeth in mammals are

- a. Acrodont. homodont and polyphiodont
- b. Acrodont. heterodont and polyphiodont
- c. Thecodont. heterodont and polyphiodont
- d. Thecodont. heterodont and diphyodont

47.

	Group	Example	Character
A	Chondrichthyes	Torpedo	Dorsal muscle modified into electric organ
B	Rhynchocephalian	Sphenodon	Copulatory organ is hemipenis
C	Urodele	Salamandra	Amphicoelous vertebrae
D	Chelonia	Testudo	Edentate

- a. A, C

- b. B, C
- c. A, D
- d. C, D

48. Match the following

List - I

- A) Isogamy
- B) Anisogamy
- C) Hologamy
- D) Conjugation

List - II

- I) Ttichonympha
- II) Euglena Plasmodium
- IV) Monocystes
- V) Paramoecium

The correct answer is

- A B C D
- a. IV III V I
A B C D
- b. IV III I V
A B C D
- c. IV II III V
A B C D
- d. III IV I V

49. The following chemical is commonly called smack.

- a. Morphine
- b. Diacetyhnoiphine
- c. Cocaine

d. Cannabinoids

50. Assertion (A) : Microfilatia exhibits nocturnal periodicity

Reason (R) : Intermediate host of Wiichereria is active only during night time

- a. Both (A) and (R) are correct and (R) is the correct explanation of (A)
- b. Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- c. (A) is correct but (R) is not correct
- d. (A) is not correct but (R) is correct

51. The following are events in life cycle of Plasmodium in female Anopheles

- a) SYllgamY
- b) Formation of gametes
- c) Oocyst formation
- d) Sporocyst formation
- e) Ookinete
- f) Zygote

Arrange them in correct sequence

- a. d, c, e, f, a, b
- b. b, a, f, c, e, d
- c. b, a, f, e, c, d
- d. b, f, a, e, c, d

52. Study the following statements regarding filarial worm

- A) The inflammation in the lymph vessels is called lymphadenitis
- B) Swelling in the extremities of limbs, scrotum and mammary glands is called lymphoedema
- C) Sausage shaped larva undergoes three moultings and transforms into infective 3rd stage microfilaria
- D) 3rd stage microfilaria larva enters the blood vessels of man and undergoes the 3rd and 4th moultings to produce young filarial worm

From the above, identify the correct statement(s)

- a. B
- b. B, C & D
- c. B & D
- d. A, B, C & D

53. Identify the mismatched pairs

- A) Trophocytes - Symbiotic bacteria
- B) Mycetocytes - Store food
- C) Urate cells - Nitrogenous wastes
- D) Oenocytes - Secrete lipids

- a. C, D
- b. A, C
- c. A, B

d. A, D

54. Match the following which are related to cockroach

List - I

- A) Adductor muscles
- B) Dorsal longitudinal muscles
- C) Alary muscles
- D) Dorsoventral muscles

List - II

- I) Depression of wings
- II) Respiration
- III) Mandibles
- IV) Excretion
- V) Blood circulation

The correct answer is

- A B C D
- a. III II V I
- A B C D
- b. I III V II
- A B C D
- c. III I V II
- A B C D
- d. IV I III II

55. If 20 joules of energy is available at the producer level as net primary productivity. then amount of energy available in secondary carnivore in joules.

A. 2

- B. 0.02
- C. 0.2
- D. 0.002

56. Assertion (A) : Circannular rhythms are controlled by biological clocks

Reason (R) : Temperature has a role in setting the biological clocks

- a. Both (A) and (R) are correct and (R) is the correct explanation of (A)
- b. Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- c. (A) is correct but (R) is not correct
- d. (A) is not correct but (R) is correct

57. Study the following statements and identify the incorrect statements.

- a. Deeper region of dermis is made up of dense irregular connective tissue
- b. Tendon is a dense regular connective tissue
- c. Pisiform bone is a visceral bone
- d. Megakaryocytes produce blood platelets

58. Fresh water fishes acquired several adaptations to withstand endosmosis problem. They are

A) Their aglomerular kidneys excrete large quantities of water through urine

B) They compensate the salt loss by absorbing salts through chloride cells

C) They retain urea in blood to keep the body fluid isotonic to aquatic medium

D) Their glomerular kidneys excrete large quantities of urine

Choose correct statements

a. A & B

b. C & D

c. B & D

d. B & C

59. Imagine the lion population of 40 in a forest cannot continue to grow exponentially due to lack of space.

The carrying capacity for their space in the forest is

80. Assume 'r' value of Lion population is 0.2

Lions/month per capita. Predict the total population in this logistic growth situation after one month.

a. 4

b. 40

c. 44

d. 84

60. The disease occurs due to deposition of iron particles in tissues is

- a. Silicosis
- b. Siderosis
- c. Emphysema
- d. Pneumonia

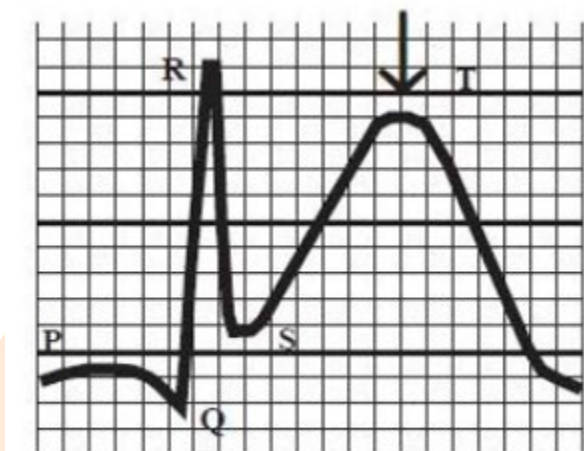
61. Study the following (consider maximum values)

- A) Inspiratory Reserve Volume (IRV)
- B) Tidal Volume (TV)
- C) Vital capacity (VC)
- D) Total Lung Capacity (TLC)
- E) Residual Volume (RV)

Arrange the above volumes of air in ascending order

- a. $B \rightarrow E \rightarrow C \rightarrow A \rightarrow D$
- b. $E \rightarrow C \rightarrow B \rightarrow A \rightarrow D$
- c. $B \rightarrow E \rightarrow A \rightarrow C \rightarrow D$
- d. $B \rightarrow E \rightarrow A \rightarrow D \rightarrow C$

62.



The above ECG indicates the following disorder

- a. Hypercalcemia
- b. Hypocalcemia
- c. Hypokalemia
- d. Hyperkalemia

63. Assertion (A): In human heart. deoxygenated blood always flows from right ventricle into pulmonary arch only

Reason (R): Mitral valve present at the base of pulmonary arch prevents the reverse flow of blood

- a. Both (A) and (R) are correct and (R) is the correct explanation of (A)
- b. Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- c. (A) is correct but (R) is not correct

d. (A) is not correct but (R) is correct

64. Study the following table

	Part of heart	Location	Function
A	Fossa ovalis	Interatrial septum	Directs blood from right atrium to left atrium
B	Sinoatrial node	Wall of right atrium	Initiation of systole
C	Atrio-ventricular node	Interventricular septum	Act as pace maker
D	Bicuspid valve	Left atrio – ventricular aperture	Directs blood to flow from left atrium to left ventricle

The correct combinations are

- a. B, C
- b. A, B
- c. A, D
- d. B, D

65. Study the following statements

- A) Atria and ventricles are externally separated by atrio-ventricular septum
- B) Two ventricles are externally separated by an oblique interventricular groove
- C) The muscular pouch like projection from each atrium is called auricular appendix
- D) In embryonic stage. the interventricular septum has fossa ovalis

Identify the correct statements

- a. A, B
- b. B, C
- c. C, D
- d. A, D

66. Identify the correct order of stages in muscle contraction

- a. Stimulus → Release of acetyl choline → release of Ca^{2+} ions → Excitation of triad system → sliding of thin filaments → Z-membranes brought closer
- b. Stimulus → Release of acetyl choline → Excitation of triad system → Release of Ca^{2+} ions

- formation of cross bridges → sliding of thin filaments → Z-membranes brought closer
- c. Stimulus → Excitation of triad system → Release of acetylcholine → Release of Ca^{2+} ions → sliding of thin filaments → formation of cross bridges → Z-membranes brought closer
- d. Stimulus → Excitation of triad system → Release of Ca^{2+} ions → Release of acetylcholine → formation of cross bridges → Z-membranes brought closer → sliding of thin filaments

67. Study the following

- A) Action potential
- B) Threshold stimulus
- C) Under shoot / Hyperpolarisation
- D) Influx of Na^+ ions
- E) Efflux of K^+ ions

Arrange the above in a correct sequence of nerve impulse generation and conduction.

- a. $\text{B} \rightarrow \text{A} \rightarrow \text{D} \rightarrow \text{C} \rightarrow \text{E}$
- b. $\text{A} \rightarrow \text{B} \rightarrow \text{C} \rightarrow \text{D} \rightarrow \text{E}$
- c. $\text{B} \rightarrow \text{D} \rightarrow \text{A} \rightarrow \text{E} \rightarrow \text{C}$
- d. $\text{B} \rightarrow \text{D} \rightarrow \text{E} \rightarrow \text{A} \rightarrow \text{D}$

68. Match the following

List – I

- A) Complement proteins
- B) Interferons
- C) Interleukins
- D) Integrase

List -II

- I) Promotes apoptosis
- II) Incorporates viral DNA
- III) Membrane attack complex
- IV) Antiviral proteins
- V) Differentiation of immune cells

The correct answer is

- A B C D
- a. III IV V II
- A B C D
- b. IV I V III
- A B C D
- c. III V IV II
- A B C D
- d. I IV V III

69. Study the following statements

- A) Calcitonin regulates the metabolism of calcium
- B) Oxytocin stimulates contraction of uterine muscles
- C) Graves disease is due to malfunctioning of adrenal gland

D) ADH stimulates water secretion and increases urine formation

Identify the incorrect statements

- a. A, B
- b. C, D
- c. A, C
- d. B, D

70. The unique feature observed in the foetus during fifth month of pregnancy

- a. Body covered with fine hair
- b. Development of limbs
- c. Appearance of hair on head
- d. Formation of heart

71. Study the following about contraception

	List – I	List – II	List – III
A	IUDS	Lippes loop	Medicated
B	OCPS	Saheli	Non steroid
C	Contraceptive injections	DMPA	Prevents pregnancy for 3 months
D	Barrier method	Vaginal ring	Release of Progesteron

E	Surgical method	Vasectomy	Removal of fallopian tube
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From the above, identify the incorrect combinations

- a. A, D & E
- b. A, C & D
- c. C, D & E
- d. D & E

72. Study the following parts of the male reproductive system.

- A) Retetestis
- B) Epididymis
- C) Urethra
- D) Seminiferous tubules
- E) Vas deferens
- F) Ejaculatory duct
- G) Vasa efferentia

The correct sequence of parts through which sperms pass out is

- a. A, B, C, D, E, F, G
- b. D, A, G, B, E, F, C
- c. D, A, G, E, B, F, C
- d. D, G, A, B, E, F, C

73. The usage of technique gel electrophoresis is
- Separation of DNA into individual bands
 - Separation of double helix DNA into single strand
 - Production of many copies of DNA
 - Cutting of DNA into small fragments
74. Assertion (A) : Klinefelter individual is Barr body positive
Reason (R): Klinefelter individual carries two X chromosomes
- Both (A) and (R) are correct and (R) is the correct explanation of (A)
 - Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 - (A) is correct but (R) is not correct
 - (A) is not correct but (R) is correct
75. Study the following statements and identify the incorrect statement
- The DNA is fragmented by restriction enzymes
 - Fragments are separated by centrifugation
 - DNA is denatured by alkali treatment or heating
 - Fragments of single strand DNA are transferred to nitrocellulose sheets and are identified with probes

76. Study the following statements and identify the incorrect statement regarding genetic drift
- a. Chances of losing a particular allele from small population is more
 - b. End result of genetic drift is either fixation or loss of allele.
 - c. Genetic drift tends to increase variations
 - d. Bottleneck effect illustrates genetic drift
77. Assertion (A) : Geographical isolation brings about the sympatric speciation
Reason (R) : If one species diverges to become two or more species, it is called cladogenesis
- a. Both (A) and (R) are correct and (R) is the correct explanation of (A)
 - b. Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 - c. (A) is correct but (R) is not correct
 - d. (A) is not correct but (R) is correct
78. Which statement is correct regarding cancer cells
- a. They exhibit contact inhibition
 - b. Cells are joined by cadherins
 - c. Cancer cells undergo apoptosis

d. They starve the normal cells by competing for
Vital nutrients

79. Match the following

List - I

- A) Sandwich assay
- B) Indirect ELISA
- C) Competitive ELISA
- D) Direct ELISA

List - II

- I) Enzyme catalysed colour production
- II) Pregnancy test
- III) Used to detect antibodies
- IV) Used to measure the amount of antigen
- V) Used to detect antigens

The correct answer is

- A B C D
- a. IV III I V
- A B C D
- b. IV III II I
- A B C D
- c. IV II III V
- A B C D
- d. IV III II V

80. Study the following statements

- A) In germ line gene therapy functional genes are introduced into somatic cells
 - B) Synthetic oligo deoxynucleosides are used for silencing the disease causing genes
 - C) The transgenic animal product α -lactalbumin is used to treat emphysema
 - D) Viral vectors are used to send genetically modified DNA by a process called transfection
- From the above. identify the correct statements

- a. A, C
- b. B, C
- c. B, D
- d. A, D

81. The sides of a rectangular plate are (9.0 ± 0.3) cm and (3.0 ± 0.1) cm. The area of the plate with error limits is

- a. $(27.0 \pm 0.1) \text{ cm}^2$
- b. $(27.0 \pm 0.3) \text{ cm}^2$
- c. $(27.0 \pm 1.8) \text{ cm}^2$
- d. $(27.0 \pm 0.2) \text{ cm}^2$

82. A particle first accelerates from rest and then retards to rest during the time interval of 8 s. If the

retardation is 3 times the acceleration. then the time for which it accelerated is

- a. 2 s
- b. 3 s
- c. 4 s
- d. 6 s

83. Two balls are projected from same place simultaneously. One ball is projected vertically upwards and the other at an angle of 30° with the horizontal. If these two reach the ground at the same time then the ratio of their initial velocities is

- a. $\sqrt{2} : \sqrt{3}$
- b. $\sqrt{2} : 1$
- c. $1 : \sqrt{3}$
- d. $1 : 2$

84. A boy can throw a ball obliquely to a maximum horizontal distance 'X' while standing on the ground. If he throws the same ball from the top of a tower of height 'r' at an angle of 45° above the horizontal from the foot of the tower. the ball hits the ground at a distance (Assume same initial speeds)

- a. $X(1 + \sqrt{2})$

b. $\frac{X}{2}(1 + \sqrt{3})$

c. $X(1 + \sqrt{4})$

d. $\frac{X}{2}(1 + \sqrt{5})$

85. A body slides down in a time 't' from rest along a smooth inclined plane making an angle of 45° with the horizontal. When the same body slides down from rest along a rough inclined plane of same length making the same angle. it takes time $1A'$ then the coefficient of friction between the body and the rough plane is (p is a constant)

a. $P^2 - 1$

b. $1 - \frac{1}{P^2}$

c. $\frac{1}{P^2}$

d. $\frac{2P^2}{1 - P^2}$

86. A car moves along a horizontal circular road of radius 'R' with a tangential acceleration ' a_t '. The coefficient of friction between the road and the tyre is ' μ '. The

maximum speed of the car to move without skidding is

- a. $\left[R^2 \mu^2 g^2 - a_t^2 \right]^{\frac{1}{4}}$
- b. $\left[R^2 \mu^2 g^2 - a_t^2 \right]^{\frac{1}{2}}$
- c. $\left[R^2 \left(\mu^2 g^2 - a_t^2 \right) \right]^{\frac{1}{2}}$
- d. $\left[R^2 \left(\mu^2 g^2 - a_t^2 \right) \right]^{\frac{1}{4}}$

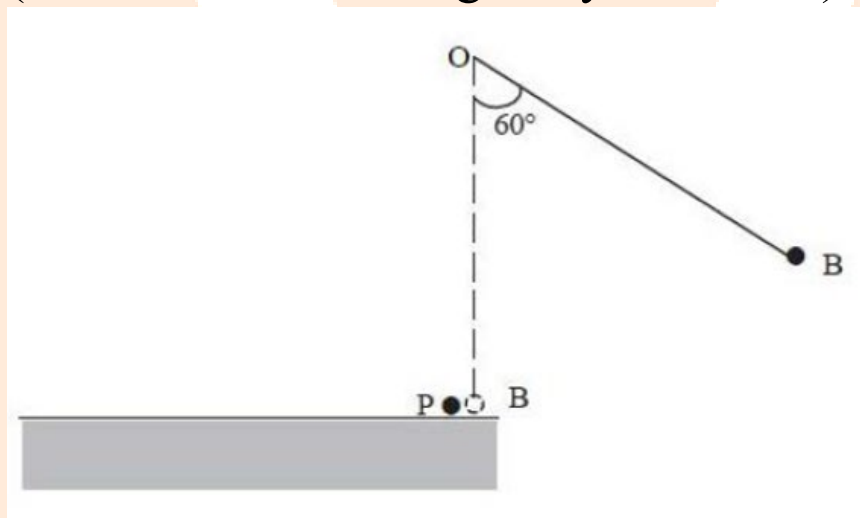
87. From a point on a smooth horizontal plane. a body is projected with a velocity 20 ms^{-1} at an angle of 45° to the horizontal. If the coefficient of restitution is 0.5. the total horizontal distance it travels before coming to rest after several rebounds is (Acceleration due to gravity = 10 ms^{-2})

- a. 40 m
- b. 60 m
- c. 80 m
- d. 120 m

88. A ball B of mass 'm' is tied to a 90 cm thread of negligible mass and is pivoted from a rigid point 'O'. The ball B is pulled aside through 60° and then

released. When the ball B reaches its lowest position. it collides elastically with another ball P of mass '2m' at rest on a rough horizontal table of coefficient of kinetic friction 0.1. as shown in the figure. After collision. the ball P travels a distance '4:1' before it comes to rest and the ball B reaches a maximum height . Then, the value of $\left(\frac{d}{h}\right)$ is

(Acceleration due to gravity = 10 ms^{-2})



- a. 40
- b. 32
- c. 20
- d. 16

89. A 15 strain ball is shot from a spring gun whose spring has a force constant of 600 Nm^{-1} . The spring is compressed by 5 cm. The greatest possible horizontal

range of the ball for this compression is (Acceleration due to gravity = 10 ms^{-2})

- a. 5m
- b. 15 m
- c. 20 m
- d. 10 m

90. Two particles P and Q are moving in a circular path of radius 5 cm. At $t = 0$, they are at the ends of a diameter and have equal angular velocity of 3 rad s^{-1} . The angular acceleration of Q is twice that of P. If both the particles move in clockwise direction they meet after 3 s. If P moves in clockwise direction and Q moves in anti-clockwise direction then they meet after 0.5 s

- a. 0.5 s
- b. 1 s
- c. 1.5 s
- d. 2 s

91. Consider a pair of identical pendulums, which oscillate with equal amplitude independently such that when one pendulum is at its extreme position making an angle of 4° to the right with the vertical, the other pendulum makes an angle of 2° to the left of

the vertical. The phase difference between the pendulums is

- a. $\frac{\pi}{4}$
- b. $\frac{2\pi}{3}$
- c. $\frac{\pi}{3}$
- d. $\frac{\pi}{2}$

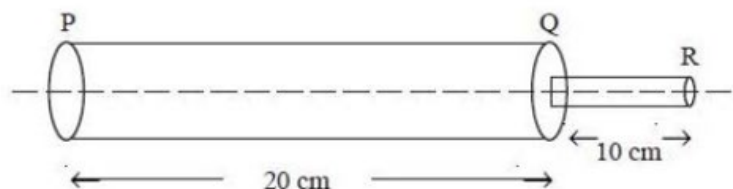
92. The time period of moon around the earth is 28 days.

If the mass of the earth is doubled. without any change in the distance of the moon from the earth. new time period of revolution of the moon is

- a. $28\sqrt{2}$ days
- b. 7 days
- c. $14\sqrt{2}$ days
- d. 14 days

93. A composite steel rod PQR is made of two rods PQ and QR as shown in figure. The lengths of two rods PQ and QR are 20 cm and 10 cm respectively. The area of cross-section of the longer rod is $2 \times 10^{-4} \text{ m}^2$ and that of the shorter rod is $1 \times 10^{-4} \text{ m}^2$. If the

composite rod is stretched with a force of $50 \times 10^3 \text{ N}$, the total elongation produced is (Young's modulus of steel = $20 \times 10^{10} \text{ Nm}^{-2}$)



- a. 5 mm
- b. 0.5 mm
- c. 25 mm
- d. 2.5 mm

94. Match the following List-I with List-II

List - I

A) Equation of continuity

B) Bernoulli's theorem

C) Turbulent flow

D) Stream line flow

List - II

I) Less than critical velocity

II) Formation of eddies and vortices

III) Law of conservation of mass

IV) Law of conservation of energy

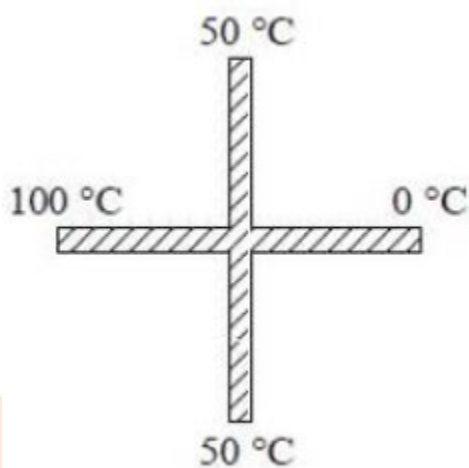
The correct answer

- A B C D
a. IV II I III
A B C D
b. III I IV II
A B C D
c. II III IV I
A B C D
d. III IV II I

95. A body cools from 60°C to 40°C in the first 7 minutes and to 28°C in the next 7 minutes. The temperature of the surroundings is

- a. 10°C
b. 20°C
c. 5°C
d. 30°C

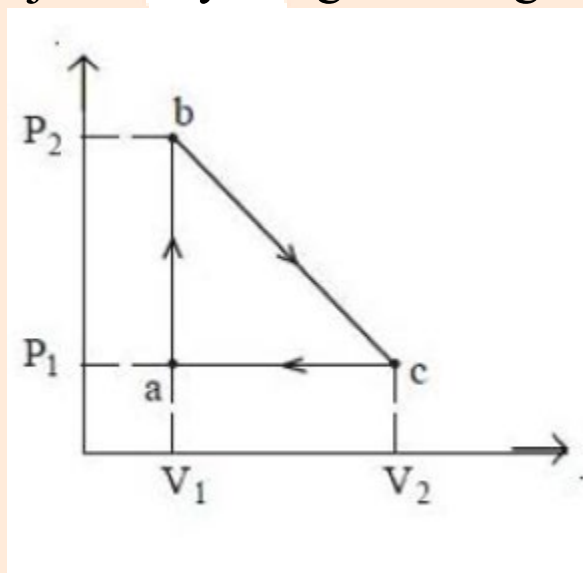
96. Two identical rods are joined as shown in the figure. The temperatures shown at the ends are in steady state. If there is no heat loss through lateral surfaces of the rods, then the temperature at the junction is



- a. 50.0 °C
- b. 75.0 °C
- c. 66.6 °C
- d. 33.3 °C

97. Two moles of a monatomic gas at 27 °C and three moles of a diatomic gas at the same temperature expand adiabatically. If the work done by each gas during the expansion is 4157 J. the ratio of the final temperatures of the monatomic gas to that of the diatomic gas is (Universal gas constant = $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$)
- a. 3:5
 - b. 4:7
 - c. 2:3
 - d. 3:7

98. Carbon monoxide $\left(\gamma = \frac{7}{5}\right)$ is carried around a closed cyclic process abc, in which 'bc' is an isothermal process as shown in figure. The gas absorbs 6000 J of heat as its temperature is increased from 200 K to 800 K in going from 'a' to 'b'. The quantity of heat ejected by the gas during the process 'ca' is



- a. 6000 J
b. 2400 J
c. 8400 J
d. 4800 J
99. Mass of each molecule of gas 'A' containing 'N' molecules is 'm' and mass of each molecule of gas 'B' containing '2N' molecules is '2m'. These two gases are contained in a vessel which is maintained at a

temperature T . The mean square velocity of the molecules of gas B is denoted by ' v^2 ' and the mean square X - component of the velocity of gas A is denoted by $\frac{W^2}{V^2}$. The value of is

- a. $\frac{4}{3}$
- b. $\frac{2}{3}$
- c. $\frac{1}{3}$
- d. $\frac{3}{4}$

100. A string fixed at both ends is vibrating in the lowest mode for which a point of maximum displacement

lies at $\left(\frac{1}{4}\right)^{th}$ of its length from one end. The

frequency of vibration is 100 Hz. If this point is to be again the point of maximum displacement, then the frequency of the next mode of vibration is

- a. 400 Hz
- b. 600 Hz
- c. 300 Hz

d. 200 Hz

101. The frequency of a note emitted by a source changes by 10% as it moves away from a stationary observer.

If it moves towards the stationary observer with the same speed, the apparent change in the frequency is

a. 10 %

b. 7.5 %

c. 12.5 %

d. 20 %

102. A double convex lens made up of a material of refractive index 1.6 is on a plane mirror. The space between the lens and the mirror is filled with water.

When a point object is placed at 30 cm above the lens system, then its image coincides with the object.

Then the focal length of the lens is

a. $\frac{20}{3}$ cm

b. $\frac{65}{3}$ cm

c. $\frac{15}{2}$ cm

d. 18 cm

103. In Young's double slit experiment, the distance between the slits is 0.5 mm and the distance between the screen and sources is 50 cm. If a light of wavelength 5000 Å is used and the total set up is immersed in a liquid of refractive index 1.5, then the fringe width is

- a. 9.3 mm
- b. 0.5 mm
- c. 3.3 mm
- d. 0.33 mm

104. Assertion (A) : When a body acquires negative charge, its mass decreases.

Reason (R) : A body acquires negative charge when it gains electrons.

- a. Both (A) and (R) are correct and (R) is the correct explanation of (A)
- b. Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- c. (A) is correct but (R) is not correct
- d. (A) is not correct but (R) is correct

105. When two point charges are at some separation in air, the force between them is F_1 . If the space between them is filled with a dielectric of dielectric

constant 4, then the force becomes F_2 . If half of the distance between them is filled with the same dielectric, then the force becomes F_3 . Then the ratio of F_1 , F_2 and F_3 is

- a. 16:9:4
- b. 9 : 36 : 16
- c. 4 : 1 : 2
- d. 36 : 9 : 16

106. Electric potential at a point $(x, 0, 0)$ is given by

$$V = \left[\frac{1000}{x} + \frac{1500}{x^2} + \frac{500}{x^3} \right] \text{ volt}$$

intensity of electric field at $x = 1\text{m}$ (in Vm^{-1}) is —5500 i . 5500i

- a. $-5500\hat{i}$
- b. $5500\hat{i}$
- c. $\sqrt{5500} \hat{i}$
- d. $-5500\hat{j}$

107. Two circular plates each of radius 'r' and charge 'q' form a parallel plate capacitor. Then the force of attraction between the plates is (Medium between the plates is air).

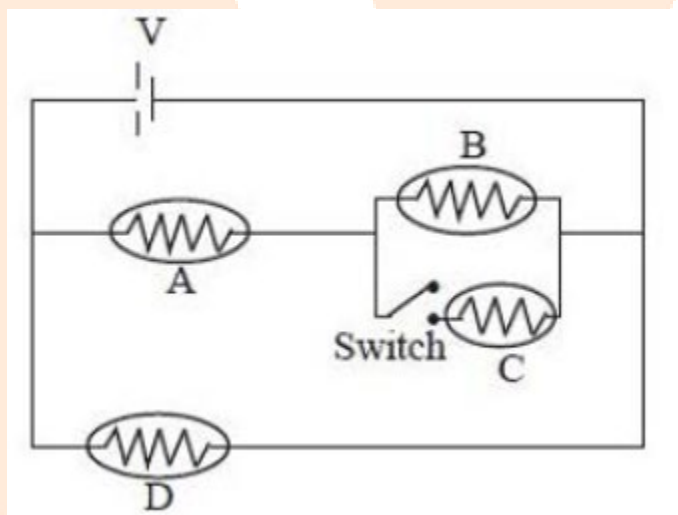
- a. $\frac{q}{2\epsilon_0 r}$

b. $\frac{q}{2\pi\epsilon_0 r^2}$

c. $\frac{2\pi\epsilon_0 r^2}{q}$

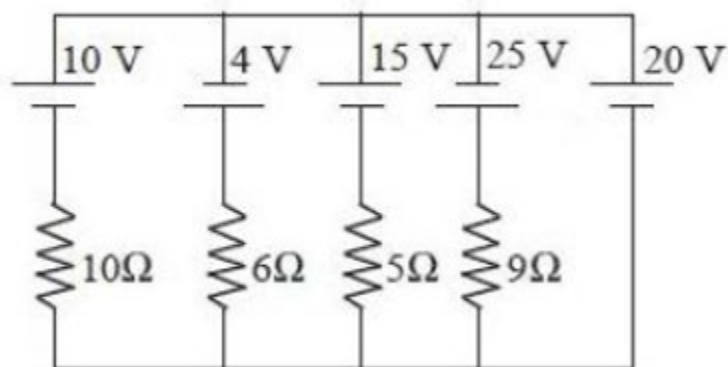
d. $\frac{q^2}{2\pi\epsilon_0 r^2}$

108. An electrical circuit consists of A. B. C. D bulbs and a battery V as shown in the figure. When the switch is closed



- a. Brightness of bulb B decreases
- b. Brightness of bulb A decreases
- c. Brightness of bulb D increases
- d. Brightness of bulb D decreases

109. In the circuit shown. the current through 20 V cell is



- a. 11 A
- b. 12 A
- c. 22 A
- d. 5.5 A

110. A current carrying conductor has 8×10^{22} free electrons per meter length having drift velocity 10^{-4} ms^{-1} . If a magnetic field of 5 T is applied perpendicular to the conductor. then the force per unit length of the conductor in Nm^{-1} is

- a. 64
- b. 3.2
- c. 16
- d. 6.4

111. An ionised gas is subjected to an electric field along positive X-axis and a magnetic field along positive Z-axis simultaneously. Then
- Positive ions deflect towards positive Y-axis and negative ions towards negative Y-axis
 - All ions deflect towards positive Y-axis
 - All ions deflect towards negative Y-axis
 - Positive ions deflect towards negative Y-axis and negative ions towards positive Y-axis
112. If the earth's magnetic field at the equator is 0.4 Ct then the earth's dipole moment is nearly (Radius of the earth = 6400 km)
- $0.73 \times 10^{22} \text{ Am}^2$
 - $1.5 \times 10^{22} \text{ Am}^2$
 - $0.87 \times 10^{22} \text{ Am}^2$
 - $1.05 \times 10^{23} \text{ Am}^2$
113. The current flowing in a coil of inductance 2 mH varies as $i = t^2 e^4$ ampere (where 't' is time in seconds). The time in which the induced emf in the coil becomes zero is
- 4s
 - 6s
 - 2s
 - 8s

114. If a direct current of 'a' units is superimposed with an alternating current $I = b \sin \omega t$, then the effective value of resulting current is

a. $I_{ms} = \left(a^2 + b^2 \right)^{\frac{1}{2}}$

b. $I_{ms} = \left(\frac{a^2}{2} + b^2 \right)^{\frac{1}{2}}$

c. $I_{ms} = \left(\frac{a^2 + b^2}{2} \right)^{\frac{1}{2}}$

d. $I_{ms} = \left(a^2 + \frac{b^2}{2} \right)^{\frac{1}{2}}$

115. The horizontal and vertical components of the electric field of an electromagnetic wave in free space are respectively $E_x = 2 \sin (\omega t + \beta z)$, $E_y = 2 \cos (\omega t + \beta z)$. Then the displacement current density is (Take $j = \sqrt{-1}$)

a. $j\omega\epsilon_0$

b. $2j\omega\epsilon_0$

c. $j\omega^2\epsilon_0$

d. $j\omega^2$

116. The ratio of de Broglie wavelengths of molecules of Hydrogen and Helium in two different gas jars at temperatures of 127 °C and 227 °C respectively is

a. $\sqrt{\frac{2}{5}}$

b. $\sqrt{\frac{5}{2}}$

c. $\sqrt{\frac{3}{5}}$

d. $\sqrt{\frac{8}{5}}$

117. If the wavelength of a photon emitted due to transition of electron from third orbit to first orbit in a hydrogen atom is X. then the wavelength of a photon emitted due to transition of electron from fourth orbit to second orbit will be

a. $\frac{22}{9} \lambda$

b. $\frac{128}{27} \lambda$

c. $\frac{36}{7} \lambda$

d. $\frac{125}{11}\lambda$

118. A radioactive element disintegrates for an interval of time equal to its mean life time. The fraction that has disintegrated is

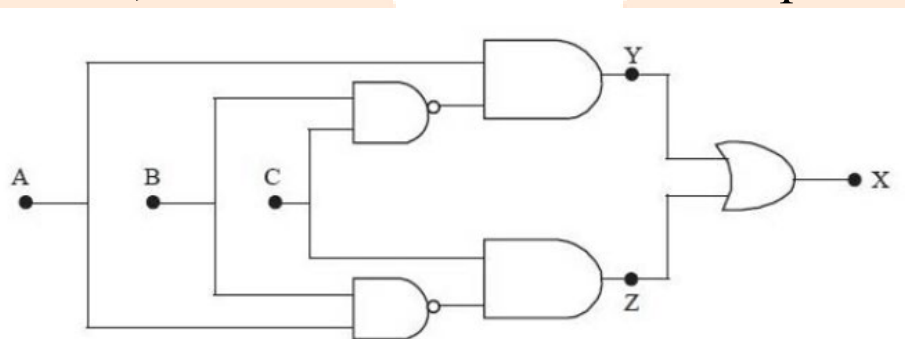
a. $\frac{1}{e}$

b. $\frac{e-1}{e}$

c. $\frac{e}{e-1}$

d. $\frac{e}{2}$

119. If $A = 1$, $B = 0$ and $C = 1$ in the following logic circuit, the values of X and Y are respectively



a. 0,1

b. 1,0

c. 1,1

d. 0,0

120. The amplitude of the modulating wave is $\left(\frac{2}{5}\right)^{th}$ of

the amplitude of the carrier wave. The percentage of modulation is

a. 20%

b. 40%

c. 50%

d. 60%

121. The energy of an electron in the excited state of hydrogen atom is $-1.36 \times 10^{-19} \text{J}$. Then according to Bohr's theory, the angular momentum of this electron in Js is

a. 4.21×10^{-34}

b. 2.11×10^{-34}

c. 1.05×10^{-34}

d. 3.16×10^{-34}

122. The wave number of the spectral line in the atomic spectrum of hydrogen is equal to $\frac{8}{9} R_H m^{-1}$. The energy associated with that spectral line is ($R_H =$ Rydberg constant)

a. $1.104 \times 10^{-24} R_H \text{ Jm}$

- b. $2.208 \times 10^{-26} R_H \text{ Jm}$
- c. $3.312 \times 10^{-27} R_H \text{ Jm}$
- d. $1.766 \times 10^{-25} R_H \text{ Jm}$

123. The oxidation state and covalency of aluminium in $[\text{Al}(\text{H}_2\text{O})_5\text{Cl}]^{2+}$ are respectively

- a. +2,3
- b. +3,
- c. +2,5
- d. +3, 6

124. The number of lone pairs of electrons present on the central atom and number of bond pairs in the following species are respectively



- | | | | |
|--------|-----|-----|-----|
| A | B | C | D |
| a. 1,4 | 1,4 | 2,2 | 0,4 |
| A | B | C | D |
| b. 2,3 | 0,3 | 1,3 | 2,2 |
| A | B | C | D |
| c. 4,1 | 0,5 | 0,2 | 4,0 |
| A | B | C | D |
| d. 2,4 | 0,4 | 1,2 | 0,4 |

125. In which of the following group of molecules, bond angles are in the increasing order?

- a. CH_4 , NH_3 , H_2O
- b. H_2O , BeCl_2 , BF_3
- c. H_2S , CH_4 , SO_2
- d. BF_3 , NH_3 , SF_6

126. Assertion (A) : In an ideal gas all the molecules travel with same velocity

Reason (R) : Ideal gas molecules neither attract nor repel each other

The correct answer is

- a. Both (A) and (R) are correct and (R) is the correct explanation of (A)
- b. Both (A) and (R) are correct and (R) is not the correct explanation of (A)
- c. (A) is correct but (R) is not correct
- d. (A) is not correct but (R) is correct

127. 50 mL of 0.1 M Na_2CO_3 (molar mass 106 g mol^{-1}) taken in a litre volumetric flask is made upto the mark by distilled water. What volume of this solution is required to exactly neutralise 20 mL of 0.02 N oxalic acid dihydrate (molar mass 126 g mol^{-1}) solution?

- a. 20 mL
- b. 30 mL
- c. 40 mL
- d. 50 mL

128. At constant temperature $T(K)$ and pressure enthalpy of combustion of ethyl alcohol is $-x \text{ J mol}^{-1}$. So. the enthalpy of combustion of ethyl alcohol at the same temperature and constant volume in J mol^{-1} will be

- a. $RT - x$
- b. $-(RT + x)$
- c. $x - RT$
- d. $x + RT$

129. Which of the following mixtures acts as a buffer solution?

- a. 0.5 L of 0.1 M CH_3COOH + 0.5 L of 0.2 M NaHCO_3
- b. 0.5 L of 0.1 M CH_3COOH + 0.5 L of 0.2 M NaOH
- c. 0.5 L of 0.1 M CH_3COOH + 0.5 L of 0.1 M HCl
- d. 0.5 L of 0.2 M CH_3COOH + 0.5 L of 0.1 M NaOH

130. The decomposition of N_2O_4 to NO_2 is carried out at $T(K)$ in chloroform. When equilibrium is reached 0.2 moles of N_2O_4 and 2×10^{-3} moles of NO_2 are present

in a 2.0 L solution. The equilibrium constant for the reaction $N_2O_4 \rightleftharpoons 2NO_2$ is

- a. 2.0×10^{-5}
- b. 2.0×10^{-4}
- c. 1.0×10^{-5}
- d. 1.0×10^{-4}

131. Which of the following is not representing the reducing property of hydrogen peroxide?

- a. $H_2O_2 + I_2 + 2OH^- \rightarrow 2I^- + 2H_2O + O_2$
- b. $2MnO_4^- + 3H_2O_2 \rightarrow 2MnO_2 + 3O_2 + 3H_2O + 3OH^-$
- c. $2MnO_4^- + 5H_2O_2 + 6H^+ \rightarrow 2Mn^{2+} + 5O_2 + 8H_2O$
- d. $PbS + 4H_2O_2 \rightarrow PbSO_4 + 4H_2O$

132. Among Li_2CO_3 , Na_2CO_3 , $ZnCO_3$, $CaCO_3$ and K_2CO_3 , the total number of carbonates which give carbon dioxide on heating is

- a. 3
- b. 2
- c. 4
- d. 5

133. In vapour phase, aluminium chloride exists as a dimer with the structure given below. The correct

relationship between the bond angles θ_1 , θ_2 and θ_3 indicated in the structure is

- a. $\theta_1 < \theta_2 < \theta_3$
- b. $\theta_3 < \theta_2 < \theta_1$
- c. $\theta_2 < \theta_1 < \theta_3$
- d. $\theta_2 < \theta_3 < \theta_1$

134. Graphite has layered structure in which each layer is made of planar hexagonal rings of carbon atoms and layers are held by van der Waals forces. The C-C bond length within the layer and distance between two adjacent layers are respectively

- a. 141.5 pm, 340 pm
- b. 340 pm, 141.5 pm
- c. 143.5 pm, 154 pm
- d. 141.5 pm, 138.3 pm

135. The reagent used in the determination of COD of polluted water is

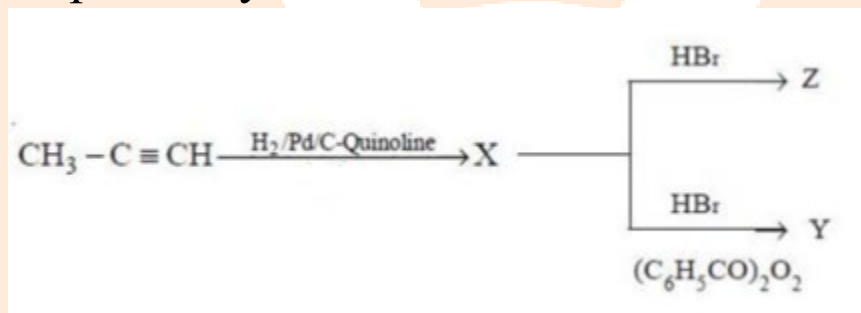
- a. 50% H_2SO_4 / $\text{K}_2\text{Cr}_2\text{O}_7$
- b. HCl / $\text{H}_2\text{C}_2\text{O}_4$
- c. HI / OH^-
- d. $\text{Na}_2\text{S}_2\text{O}_3$ / H^+

136. In the set of species

HS^- , BF_3 , R_3N , $\overset{\oplus}{N}O_2AlCl_3$, H_2O , $C_2H_5O^-$. the number of electrophiles and nucleophiles are respectively

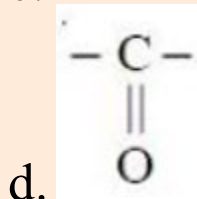
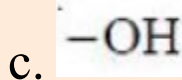
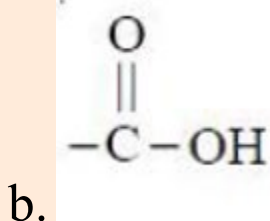
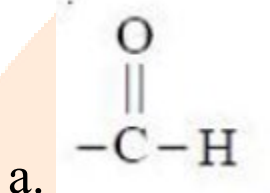
- a. 3, 4
- b. 4, 3
- c. 2, 5
- d. 5, 2

137. In the formation of Y and Z as major products. in the following reactions, the intermediates involved respectively are



- a. $H_3C-CH_2-\overset{\oplus}{C}H_2$, $CH_2-\overset{\bullet}{C}H-CH_2-Br$
- b. $CH_3-\overset{\oplus}{C}H-CH_3$, $CH_3-\underset{\underset{Br}{|}}{CH}-CH_2$
- c. $CH_3-\overset{\bullet}{C}H-CH_2-Br$, $CH_3-CH_2-\overset{\oplus}{C}H_2$
- d. $CH_3-\overset{\bullet}{C}H-CH_2-Br$, $CH_3-\overset{\oplus}{C}H-CH_3$

138. The homologue of ethyne on addition of water in the presence of Hg^{2+} & H^+ at 333 K gives an unstable compound X. This on isomerisation (tautomerisation) gives Y. The functional group in Y is



139. The percentages of Ni^{2+} and Ni^{3+} present in nickel oxide having the formula $\text{Ni}_0.9\text{S}_0$ are respectively

- a. 95.9, 4.1
- b. 4.1, 95.9
- c. 98.0, 2.0
- d. 2.0, 98.0

140. At $T(K)$, the vapour pressure of a pure solvent X is 0.8 atmospheres. What is the mole fraction of the non-volatile solute Y in the solution, which drops the vapour pressure of X to 0.6 atm?

- a. 0.5
- b. 0.75
- c. 0.25
- d. 0.65

141. 2.5 g of phenol (molar mass 94 g mol^{-1}) dissolved in 200 g of benzene shows a depression in freezing point equal to 0.512 K. What is the percentage of association of phenol if it forms a dimer in solution? (K_f of benzene = $5.12 \text{ K kg mol}^{-1}$)

- a. 50.4
- b. 49.6
- c. 73.2
- d. 26.8

142. If the resistance of the conductivity cell filled with 0.02 M KCl solution is 516Ω , the molar conductivity of the same solution in $S \text{ cm}^2 \text{ mol}^{-1}$ is (cell constant = 129 m^{-1})

- a. 1.25×10^{-2}
- b. 2.50×10^2

c. 1.25×10^{-3}

d. 2.50×10^3

143. If the time required for the completion of 75% of a first order reaction is 30 minutes. the time required for the completion of 93.75% of the reaction in minutes is

a. 45

b. 30

c. 60

d. 75

144. In the coagulation of $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ sol. the coagulating power of the anions follow the order

a. Phosphate > Chloride > Sulphate

b. Chloride > Phosphate > Sulphate

c. Sulphate > Phosphate > Chloride

d. Phosphate > Sulphate > Chloride

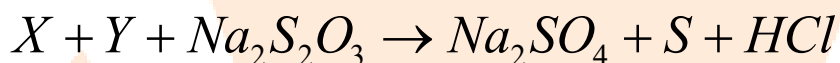
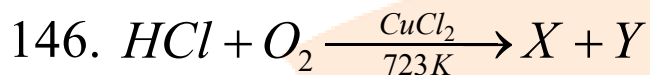
145. The number of carbonate. sulphide ores respectively from the ores given below is: Calamine. Copper glance. Fool's gold. Sphalerite. Siderite. Chalcopyrites

a. 2, 4

b. 3, 3

c. 4, 2

d. 1, 5



X and Y in the above reactions are

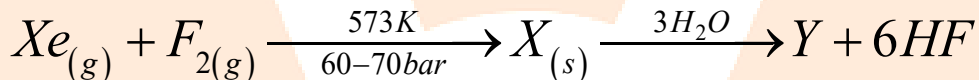
a. Cl_2O , H_2O

b. H_2O , Cl_2

c. H_2O_2 , ClO_2

d. Cl_2 , H_2O_2

147. If Xe and F_2 are in 1:20 molar ratio in the reaction



The shapes of X and Y are respectively

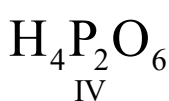
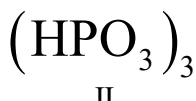
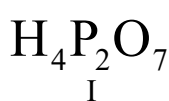
a. Distorted Octahedral, Pyramidal

b. Pyramidal, Distorted Octahedral

c. Square Pyramidal. Pyramidal

d. Pyramidal, Square Planar

148. The number of P-O-P bonds present in the oxoacids of phosphorous given below are respectively



I II III IV

- a. 2 1 0 1
I II III IV
- b. 1 3 1 0
I II III IV
- c. 1 2 1 0
I II III IV
- d. 0 2 2 1

149. The number of moles of dichromate ($Cr_2O_7^{2-}$) and permanganate (MnO_4^-) separately required to oxidise 1 mole of ferrous iron each in acidic medium are respectively

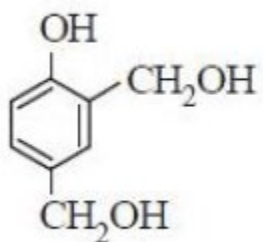
- a. $\frac{1}{6}, \frac{1}{5}$
- b. $\frac{1}{5}, \frac{1}{6}$
- c. $\frac{1}{4}, \frac{1}{5}$
- d. $\frac{1}{2}, \frac{1}{2}$

150. The Wilkinson catalyst used in the hydrogenation of alkenes is

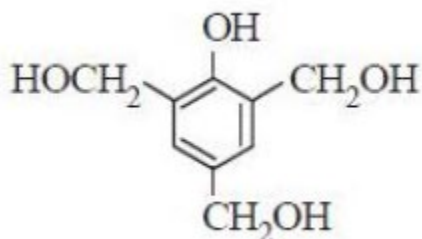
- a. $Na_3 [Ag (S_2O_3)_2]$

- b. $[(\text{Ph}_3\text{P})_3\text{RhCl}]$
- c. $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
- d. $[\text{Ni}(\text{CO})_4]$

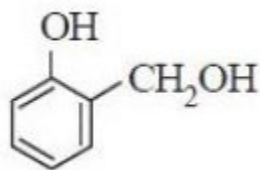
151. Novolac, a linear polymer is formed from which one of the following?



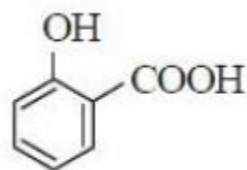
a.



b.



c.



d.

152. The number of $-CH_2OH$ groups and $-CHOH$ groups present in the open chain structures of D-(+)-glucose (A) and D-(—)-fructose (B) are respectively

A B

a. 0,5 1,4

A B

b. 2,3 1,4

A B

c. 1,4 2,3

A B

d. 3,2 4,0

153. Which one of the following artificial sweetners does not have $—CONH—$ group?

a. Aspartame

b. Sucralose

c. Alitane

d. Saccharin

154. Which of the following statements are correct with respect to S_N2 reactions?

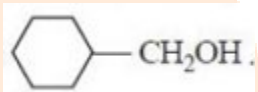
I) They follow second order kinetics

II) Optically active alkyl halides give racemic products

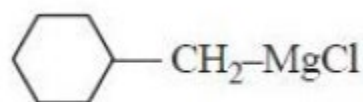
III) In the case of optically active alkyl halides, the product formed has inverted configuration compared to the reactant IV) These reactions proceed through intermediates

- a. I, III, IV
- b. I, III only
- c. II, IV only
- d. I, IV only

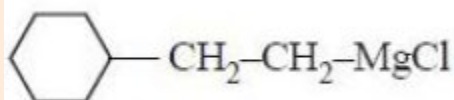
155. Methanol on reaction with Grignard reagent A followed by hydrolysis in the presence of acid gives



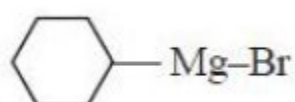
A is



a.



b.



c.



d.

156. Identify the set from the following, that produces anisole

- $\text{C}_6\text{H}_5\text{CHO}$, CH_3MgCl
- $\text{C}_6\text{H}_5\text{OH}$, NaOH , CH_3I
- $\text{C}_6\text{H}_5\text{OH}$, NaOH , CO_2
- $\text{C}_6\text{H}_5\text{COCH}_3$, LiAlH_4

157. Match the following

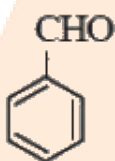
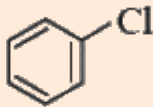
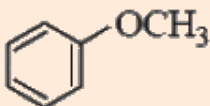
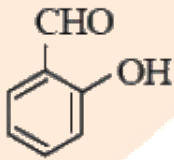
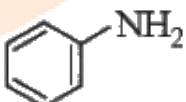
List - I

(Name Reaction)

- Reimer-Tiemann
- Hoffmann bromamide
- Rosenmund
- Sandmeyer

List - II

(Main organic product)

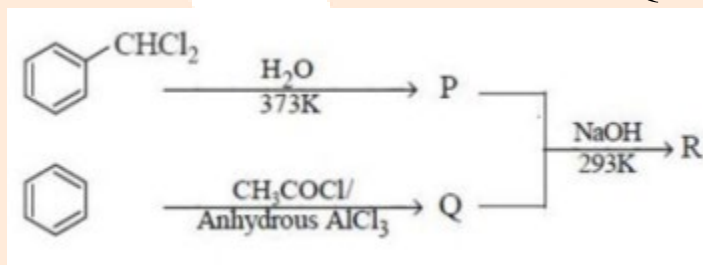
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- 
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- 

The correct answer is

A B C D

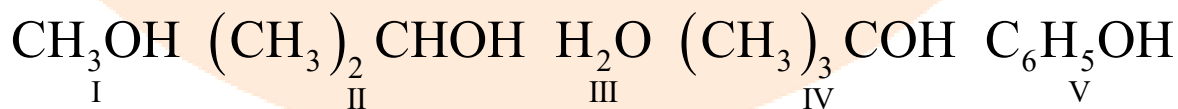
- a. V III I II
A B C D
- b. IV V I II
A B C D
- c. IV V III I
A B C D
- d. III IV II V

158. Formation of R from P and Q is an example of



- a. Cannizaro reaction
 b. Simple aldol condensation
 c. Mixed aldol condensation
 d. Reimer - Tiernan reaction

159. Arrange the following in the order of their acidic strength



- a. V > III > I > II > IV
 b. III > V II > I > IV
 c. IV > II > I > III > V

d. I > II > IV > V > III

